Security of Retrieval-Augmented Generation Against Targeted Knowledge-Base Poisoning

An exploratory offline study of retrieval and extractive answer selection

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Abstract

Targeted knowledge-base poisoning can place an attacker-selected false claim alongside relevant evidence. This exploratory Afghanistan-focused study distinguishes supporting-evidence recovery, malicious exposure, targeted answer corruption, and correct-answer restoration. The archived principal experiment uses 224 documents, 225 chunks, 364 questions, 104 targets, and 21,840 correlated question-condition evaluations. Its actual pipeline comprises 1,024-dimensional hashing, exact cosine retrieval, TF-IDF reranking, and deterministic sentence extraction; its numerical results do not measure generative LLM behavior. A local audit recomputed response scores and replayed the full grid with checksum-matched historical code. At full insertion, gold Recall@5 falls from 93.4% to 5.8%, lexical answer-value accuracy from 82.4% to 3.6%, and exposure-confirmed lexical attack success reaches 73.1%. Reranking restores Recall@5 to 88.7% but leaves attack success at 72.1%; 73 targets retain gold evidence while producing the intended false value. Eligibility filtering prevents rejected-label attacks. A separate target-specific accepted-label control yields 91/104 successes, demonstrating dependence on metadata authority rather than implemented source authentication. Separately archived neural retrieval controls and a 24-response generative development pilot demonstrate implementation progress, not held-out generalization or human-adjudicated security rates. The contribution is a bounded empirical analysis of distinct security outcomes. Fresh reference-checked data, independent semantic review, and a frozen final generative experiment remain necessary for broader claims.

Keywords: retrieval-augmented generation; knowledge-base poisoning; retrieval integrity; answer integrity; provenance; adversarial evaluation

1. Introduction

Retrieval-augmented generation combines a language model with external evidence to support knowledge-intensive tasks [11–13]. In institutional applications, retrieved material may include public reports, administrative records, policy documents, or statistical observations. This access makes the integrity of the knowledge base a security concern: a document can be highly relevant to a question while supplying a false answer. A retrieval score consequently indicates relevance under a scoring rule, not the factual reliability or authorized origin of the retrieved claim.

Prior research demonstrates that corpus insertions can influence downstream RAG responses [1–4]. The resulting security question extends beyond whether malicious text can be retrieved. A useful assessment must distinguish whether supporting evidence remains available, whether attacker-controlled material enters the answer context, and whether the answer adopts the intended false claim. These outcomes can diverge. A defense that improves retrieval may preserve both correct and false evidence, leaving the decision between them unresolved.

This study investigates those distinctions in an Afghanistan-focused factual corpus. The practical implementation is the experimental instrument. The principal results describe an offline retrieval-and-extraction pipeline whose deterministic behavior permits controlled comparisons. They must not be attributed to generative language models. Appendix D separately records subsequent neural and generative development checks; those checks do not replace the historical design or establish held-out performance.

The research makes three bounded contributions. First, it evaluates supporting-evidence retrieval, malicious exposure, answer-value correctness, and target-specific success within one controlled design. Second, paired outcomes distinguish attack prevention from correctness restoration and newly introduced failures. Third, separate repetition and forged-label controls identify the limits of duplicate handling and metadata authority. The archived grid is accompanied by a local response-level audit and re-execution. These contributions constitute an exploratory domain study and analytical synthesis; no new attack algorithm, certified defense, or representative evaluation of deployed generative RAG is claimed.

For institutions considering knowledge-based assistants, the resulting distinction is consequential: recovering the correct source is an intermediate property, while producing a reliable answer is the final objective. This paper examines that relationship and identifies the conditions under which the studied controls offer protection.

2. Related research and study positioning

2.1. Retrieval augmentation and evidence use

RAG and dense passage retrieval establish the architectural basis for evidence-grounded question answering [11, 12]. REPLUG extends retrieval augmentation to black-box language models [13]. These approaches motivate separate assessment of evidence selection and response construction. Work on long-context use also shows that a language model's use of evidence can depend on its placement [16]. The present extractive experiment does not reproduce those model-level effects; it examines the more limited relationship between context membership and selected answer values.

2.2. Corpus poisoning and adjacent threats

PoisonedRAG studies targeted knowledge corruption through malicious knowledge-base insertions [1]. CorruptRAG investigates practical attacks using a single poisoned text [2]. Su et al. examine adversarial context and retrieval robustness [3], while a broader benchmark compares multiple poisoning attacks and defenses [4]. These studies establish that targeted factual corruption is an existing research problem. Our contribution concerns the interpretation of defense outcomes within a bounded factual domain.

Attacker capabilities must be distinguished when relating results. PR-Attack coordinates poisoned content with a prompt trigger [5], whereas the base attacker here controls corpus insertions only. PANDORA studies poisoning for jailbreak objectives [6]. Indirect prompt injection uses external content to convey instructions [14]; privacy attacks pursue information disclosure [7]; and instruction-tuning poisoning changes training data [15]. These threats share an interest in untrusted information but differ in access, intervention, and success criteria.

2.3. Defenses and outcome measurement

RobustRAG uses an isolate-then-aggregate strategy and develops guarantees under a defined retrieval-corruption model [8]. RAGPart and RAGMask examine retrieval-stage defenses [9]. The

present lexical reranker and metadata gate implement neither those methods nor their guarantees. They are practical comparison conditions whose effects must be interpreted relative to the attacker and the provenance authority.

Natural Questions and HotpotQA provide potential independent question-answering settings [17, 18]. BEIR motivates heterogeneous retrieval evaluation [19]. RAGAs and citation-based answer evaluation address complementary dimensions of response quality and grounding [20, 21]. They do not remove the need to identify whether a particular attacker-selected false answer was produced. Table 1 positions the study by objective and threat boundary; it is not a cross-paper numerical ranking.

Table 1. Selected research positioning. Descriptions summarize the cited research; methods are not claimed to have been reproduced.

Research	Principal focus	Relation to this study
PoisonedRAG [1]	Targeted knowledge corruption in generative RAG	Establishes the corpus-insertion threat; this pilot uses an extractive answerer
CorruptRAG [2]	Practical single-text poisoning	Motivates separating insertion count from per-target strength
Su et al. [3]	Adversarial context and retrieval robustness	Supports separate retrieval and downstream assessment
Benchmarking poisoning [4]	Multiple attack and defense comparisons	Provides broader coverage than the present domain study
RobustRAG [8]	Certified robustness under retrieval corruption	Uses a different defense and guarantee model
RAGPart and RAGMask [9]	Retrieval-stage defenses	Provides advanced defense context; not implemented here
Present study	Evidence recovery, paired answer outcomes, and provenance assumptions	Controlled offline pilot with explicit limits on generative conclusions

3. Analytical framework and research questions

3.1. Evidence availability, exposure, and answer integrity

Let K be the clean knowledge base and P_B a set of B inserted malicious passages. For query q and defense d, the pipeline produces final context $T_d(q, K \cup P_B)$ and answer $A_d(q)$. Let G(q) indicate the annotated supporting chunk in that context, R(q) the target question's own malicious passage, C(q) a lexical match to the expected value, and F(q) a lexical match to its designated false value. Let Z(q) identify designated targets. Matching indicators measure text containment under normalization and token boundaries; they do not establish semantic correctness.

The scored answer-value correctness and targeted-success indicators are:

$$V(q) = C(q) \times [1 - F(q)]; \quad S(q) = Z(q) \times R(q) \times F(q) \times [1 - C(q)].$$

For questions without an assigned false value, F(q) is zero. An answer matching both values is flagged as ambiguous and receives neither correctness credit nor attack-success credit. No such ambiguous answers occur in the archived main run, but the exclusion remains part of the scoring rule.

Mixed or negated claims can still be unsafe despite this conservative convention. Target-specific retrieval is necessary for credited attack success; another question's poison does not satisfy that requirement.

The central analytical distinction follows from the pipeline: $G(q) = 1$ does not imply $R(q) = 0$, and neither context membership indicator determines the output value. Correct and malicious passages can coexist. Thus high gold recall cannot, on its own, establish resistance to targeted manipulation. Conversely, eliminating malicious context can restore clean behavior without correcting errors already present in the clean answerer.

3.2. Attack coverage and interpretation of dose curves

Let $I_B(q)$ indicate whether the passage targeting q is inserted. The shared-corpus design inserts one passage per selected target, and the scored outcome requires retrieval of that target's own passage. Therefore $S(q) \leq I_B(q)$. With 104 designated targets and B distinct target passages, the all-target success rate satisfies:

$$ASR_all(B) = \Sigma\, S(q) / 104 \leq B / 104.$$

This is a counting bound derived from the experiment's success definition, not a robustness guarantee. At $B = 10$, all-target ASR cannot exceed approximately 9.6% under this design, even if every inserted attack succeeds. An increasing ASR curve can consequently reflect wider coverage as well as changes in retrieval competition. To isolate per-target strength, the target set must remain fixed while attack multiplicity or diversity changes. The supplementary repetition experiment addresses multiplicity under a different protocol; it is analyzed separately.

3.3. Research questions and exploratory expectations

RQ1 examines how poisoning changes supporting-evidence retrieval and malicious exposure. RQ2 examines the relationship between exposure, answer-value correctness, and intended false answers. RQ3 examines the protection provided by reranking, trusted-label filtering, and their combination. RQ4 examines the corresponding clean-performance cost.

Four exploratory expectations are examined within the historical design: H2, poisoning is associated with increased incorrect or manipulated answers; H3, reranking reduces malicious rank or inclusion; H4, filtering with reranking provides greater protection than reranking alone; and H5, defenses may impose a clean-performance cost. Their original identifiers are retained for traceability; these expectations were not preregistered. H1, an advantage of targeted over matched unrelated poison, is an untested prospective question because the required comparison condition is absent. It is excluded from claims of empirical support.

3.4. Attacker capabilities and trusted authority

The base attacker inserts passages into a separate copy of the clean corpus. Each passage associates a designated question with a selected false value. Existing clean text, retrieval and answerer code, and defense implementation remain fixed. Prompt control, training-data modification, and production-system compromise are outside this base condition. The study does not establish performance against adaptive attack optimization.

The metadata condition assumes that an ingestion authority assigns reliable trust and verification labels. All 104 base attack entries carry rejected labels: untrusted and unverified. Of the 225 clean chunks, 223 are eligible; two inherited mineral-resource chunks are verified but have low trust and are rejected. The filter implements an eligibility policy over supplied labels, without independently

establishing source authenticity or factual truth. A separate forged-label control marks malicious entries high and verified, changing the attacker's capability. The two trust boundaries are analyzed separately.

4. Materials and experimental method

4.1. Corpus and reference questions

The corpus combines 56 inherited public-information documents and 168 World Development Indicators observations across 12 Afghanistan indicator families for 2010–2023 [10]. It contains 224 documents and 225 chunks. The question set contains 28 inherited questions and 336 indicator questions, with two formulations for each indicator-year observation. There are 104 designated targets and one original malicious passage per target. Table 2 summarizes the composition.

Table 2. Composition of the experimental material.

Component	Public-information material	Indicator material	Total
Documents	56	168	224
Questions	28	336	364
Attack targets	20	84	104
Original target passages	20	84	104

The 225 chunks form the complete clean retrieval collection. All 224 documents have indexed text. The 168 World Bank observations were checked against stored API response bodies and their recorded SHA-256 hashes. Each has two question formulations, giving 336 statistical questions; values are rounded to two decimals with decimal half-up rounding, without imputing missing observations. The 56 inherited documents retain verification labels that have not been independently re-audited. Related indicator-years and question formulations are correlated. Indicator-level development, validation, and test partitions exist, but were inspected during development; this all-question study therefore claims no untouched test set. The complete clean KB remains visible for every partition. The English question set does not establish capability in Dari or Pashto.

4.2. Executed retrieval-and-extraction pipeline

The principal experiment is the archived run of 21 September 2026, 20260921T061151Z_d1966106. Its manifest, response records, configuration, and checksum-matched source identify 1,024-dimensional token hashing, deterministic exact cosine search, a query-and-candidate TF-IDF reranker, and an extractive-stub answerer. The retriever produces 30 candidates; the final context contains at most five passages. The historical answerer scans sentences in context order and selects the sentence with greatest query-keyword overlap, retaining the first sentence on ties. It excludes a fixed stopword list, including Afghanistan, and does not reason about claim entailment.

The extractive backend ignores the configured system prompt, temperature, and maximum output-token parameter. Their presence in a configuration file is consequently not evidence of instruction-following protection or controlled LLM generation. Neural embedding and cross-encoder model names also appear in configuration fields, but the recorded active backends are hashing and TF-IDF. Later lexical/evidence revisions change both ranking and sentence selection; Appendix B documents their separate diagnostic.

Filtering accepts passages labelled verified with high or medium trust. It implements an eligibility rule over supplied metadata; it does not authenticate a publisher, validate a document signature or determine whether a claim is true. Eligibility precedes candidate retrieval; the combined condition then reranks eligible candidates. This order can prevent rejected entries from exhausting the candidate pool. It differs from reranking all candidates before filtering, which can leave an undersized context. Figure 1 shows the implemented conditions under the assumed label authority.

Table 3. Executed settings for the principal pilot.

Setting	Recorded configuration
Clean collection	225 chunks
Retriever	1,024-dimensional hashing; exact cosine similarity
Candidate count	30
Maximum final context	5 passages
Reranker	TF-IDF cosine similarity fitted to query and candidates
Answerer	Deterministic lexical extraction (`extractive-stub`)
Filter rule	Verified and high/medium trust; before retrieval
Shared-corpus budgets	0, 10, 30, 60, 104 inserted passages
Selection seeds	17, 42, 91
Defense settings	None; reranking; eligibility filtering; filtering then reranking

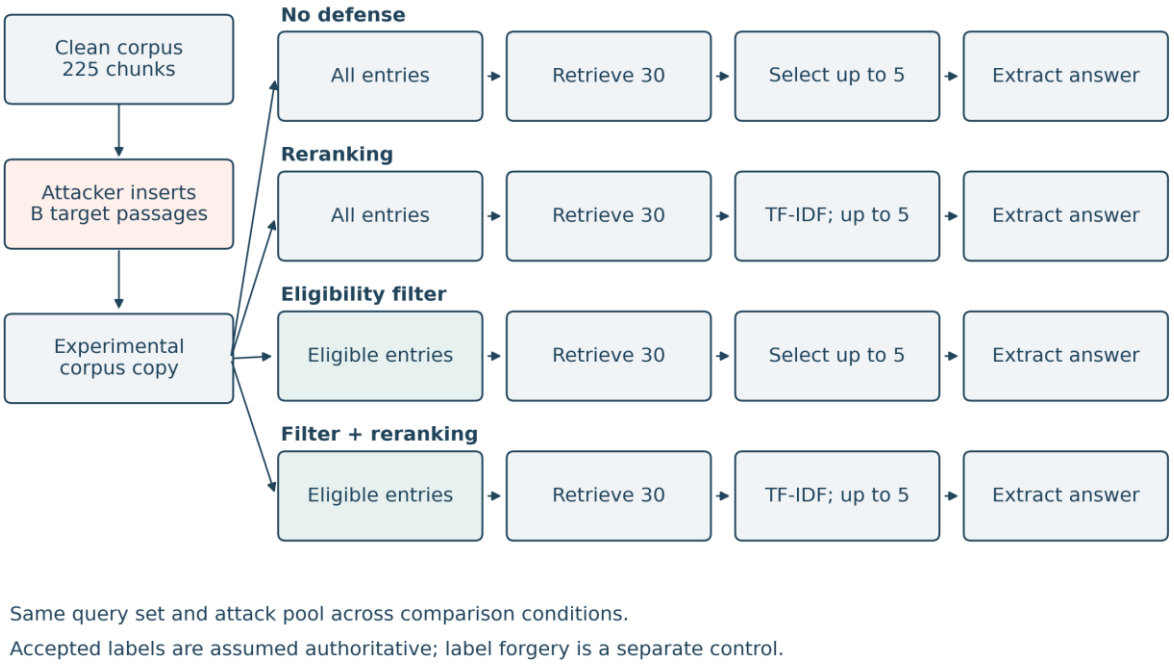


Figure 1. Conceptual view of the executed pilot. The four conditions share the same clean collection, insertion pool, and extractive answerer. Eligibility precedes retrieval where enabled. Attacker access is limited to corpus insertion in the base experiment; label forgery is a separate control.

4.3. Experimental comparisons

Each of 364 questions is evaluated at insertion budgets 0, 10, 30, 60, and 104, under four defense settings and three attack-selection seeds (17, 42, 91): 364 × 5 × 4 × 3 = 21,840 evaluations. The corresponding inserted shares are 0.0%, 4.3%, 11.8%, 21.1%, and 31.6%, calculated as B/(225 + B). Each malicious passage is indexed as one preconstructed chunk; the budgets count passages rather than independently sampled documents or sources. A seeded permutation selects nested subsets at intermediate budgets. At full insertion, 104/225 = 46.2% is the amount added relative to the clean corpus, whereas 104/329 = 31.6% is the malicious share of the resulting corpus. This severe stress condition does not estimate institutional prevalence or performance at 1% contamination.

Seeds select reproducible subsets from the fixed attack pool. They do not sample independent generated responses. At zero and full budget, selections are identical across seeds. Intermediate budgets change both target coverage and shared-corpus competition, as explained in Section 3.2.

4.4. Outcomes and denominators

Gold Recall@5 is the fraction of all 364 questions whose single annotated supporting chunk appears in the final context. MRR averages its reciprocal final-context rank, with zero when absent. These annotations may omit alternative legitimate support. Answer-value accuracy averages V(q) over the same 364 questions. Target-specific exposure and all-target ASR use the 104 designated targets. ASR among injected targets uses only questions whose own attack is present in the KB: the denominator equals B at each positive budget in this all-question protocol. It is undefined at B = 0 because no target has an inserted attack. All-designated-target ASR is zero at that budget under the exposure-confirmed rule; that zero is not evidence of protection. Any-poison exposure uses all 364 questions; malicious occupancy is the fraction of final context slots containing any inserted poison.

Answer-value scoring lowercases and normalizes text, strips selected hedge prefixes and parenthetical material from reference values, removes commas and currency symbols, and uses token boundaries. It excludes simultaneous expected/false-value matches from accuracy. Strict normalized exact match and token F1 are auxiliary measures. Value containment cannot reliably distinguish negation, quotation, the wrong entity, the wrong year, or incompatible units. The metric is therefore lexical answer-value accuracy, not fully adjudicated factual or semantic accuracy.

For defense d, let U be the targets with successful undefended attacks at a fixed budget. Paired prevention counts targets in U whose defended attacks no longer succeed, divided by |U|. Correctness restoration counts targets in U with correct defended answers, using the same denominator. Newly successful attacks count targets outside U whose defended outputs become scored successes. Clean cost is the clean accuracy without defense minus clean accuracy with defense, in percentage points. These quantities describe different transitions and should not be substituted for one another.

4.5. Analysis and supplementary protocol

Across-seed means and sample standard deviations describe sensitivity to attack selection. At zero and full budget, the attack sets, contexts, and answers are identical across seeds. These repeated rows are not independent samples, and no confirmatory p-values or population confidence intervals are claimed. The archived software exports descriptive Wilson intervals; the article does not interpret them as inferential uncertainty. Figures are generated from unrounded per-question proportions rather than from rounded summary percentages.

Evidence verification used a separate analysis script to check archived output hashes, current input hashes, question-to-reference links, attack selections, retrieved text and metadata, and all five lexical verdict fields. All 21,840 main records were rescored without disagreement. The full grid was then

locally re-executed using all 32 source files recovered exactly to the main manifest's recorded hashes. Answers, retrieved IDs and rounded scores, exposure indicators, lexical verdicts, and token F1 matched the archive for every record. Latency was excluded from equivalence claims. This is local reproducibility evidence, not an external replication or semantic adjudication; Appendix C records the verification scope.

The supplementary protocol evaluates the 104 targets separately at 0, 1, 2, 5, 10, 20, and 50 identical copies, under no defense, reranking, metadata filtering, and exact-text duplicate collapse. It contains 2,912 evaluations, plus 104 forged-label evaluations at 50 copies, for 3,016 in total. This target-specific intervention differs from adding distinct attacks to a shared corpus, so its outcomes are not pooled with the factorial experiment. A later revised-backend diagnostic is documented separately in Appendix B.

5. Results

5.1. Supporting evidence and malicious exposure

At full insertion, undefended gold Recall@5 falls from 93.4% to 5.8%, while MRR falls from 0.837 to 0.045. The target's own poison appears in the final context for 97 of 104 targets. Any poison appears for 97.5% of all questions, and malicious passages occupy 93.0% of final context slots on average. Poisoning therefore affects both directly targeted questions and the broader availability of useful evidence.

Reranking restores full-budget gold Recall@5 to 88.7% and MRR to 0.637, yet own-poison exposure rises from 97 to 98 of 104 targets. In 73 target questions, the final reranked context contains the gold chunk and the target poison while the answer meets the false-only attack-success rule. This observed coexistence directly demonstrates that evidence recovery does not establish answer integrity. H1 remains untested because there is no matched unrelated-poison condition.

Table 4. Principal pilot endpoints. Recall and answer-value accuracy use 364 questions; own-poison exposure and ASR use 104 targets. Values are percentages rounded to one decimal place. Endpoints are identical across selection seeds.

Budget	Defense	Recall@5	Own-poison exposure	Answer-value accuracy	ASR
0	None	93.4	0.0	82.4	0.0
0	Reranking	98.1	0.0	81.3	0.0
0	Eligibility filter	93.4	0.0	82.4	0.0
0	Filter then reranking	98.1	0.0	81.3	0.0
104	None	5.8	93.3	3.6	73.1
104	Reranking	88.7	94.2	42.0	72.1
104	Eligibility filter	93.4	0.0	82.4	0.0
104	Filter then reranking	98.1	0.0	81.3	0.0

5.2. Exposure and intended answer corruption

Undefended all-target ASR averages 9.0%, 23.1%, 46.8%, and 73.1% at budgets 10, 30, 60, and 104. Sample standard deviations across selection seeds are 0.6, 1.0, 1.1, and 0.0 percentage points. Among

injected targets, the corresponding mean ASRs are 93.3%, 80.0%, 81.1%, and 73.1%. The increasing all-target curve consequently combines increasing coverage with changing corpus competition; it does not show increasing susceptibility of a fixed attacked population. Both denominators are retained in the reproducibility tables.

At full budget, 76 attacks succeed among 104 designated targets, compared with 97 targets exposed to their own poison. Exposure therefore does not guarantee scored success even for this simple answerer. Overall answer-value accuracy falls from 300/364 clean answers to 13/364. This damage extends beyond the 76 intended manipulations: non-targeted questions can lose supporting evidence or retrieve another target's poison.

Among the 83 targets answered correctly in the clean condition, 73 suffer a scored targeted success at full insertion, or 88.0%. This paired restriction distinguishes newly induced failures from inherited clean errors. The results are consistent with H2 for the implemented answerer, but do not establish its magnitude for generative LLMs.

Strict normalized exact match is zero throughout the factorial run because extracted sentences do not exactly match the short reference strings. Clean token F1 is 0.093 without defense, decreasing to 0.006 at full insertion; reranking yields 0.050 at that budget. These auxiliary results chiefly expose output/reference format sensitivity and do not independently validate semantic correctness.

5.3. Prevention, restoration, and newly induced failures

At full budget, reranking raises overall answer-value accuracy from 3.6% to 42.0%, but targeted successes decline only from 76 to 75. Paired analysis explains the distinction: two undefended successes cease to succeed, one previously unsuccessful attack becomes successful, and only one of the 76 initially successful attacks receives a correct defended answer. Paired prevention is 2.6%; correctness restoration is 1.3%. Table 5 retains these transitions rather than representing the net ASR change as comprehensive protection.

Table 5. Paired defense outcomes at budget 104. Prevention and restoration use the 76 undefended successful attacks. A single endpoint seed is sufficient because the inserted pool is identical across seeds.

Defense	Prevented successes	Correct answers restored	Newly successful attacks
Reranking	2/76	1/76	1
Eligibility filter	76/76	73/76	0
Filter then reranking	76/76	73/76	0

Filtering eliminates own-poison exposure and scored success at every budget, recovering the corresponding clean pipeline behavior. It prevents all 76 full-budget successes but restores correctness for only 73 because some targets remain wrong even without poisoning. The combined condition provides stronger protection than reranking alone under the authoritative-label assumption, consistent with a restricted interpretation of H4. It does not improve ASR over filtering alone. The ablation attributes the prevention effect to eligibility rather than to an additional protective interaction with reranking.

H3 receives mixed support. Reranking reduces overall malicious occupancy and recovers annotated support, yet it does not reduce full-budget own-poison inclusion. Relevance discrimination and factual discrimination are distinct properties. These observations do not establish the behavior of a learned cross-encoder.

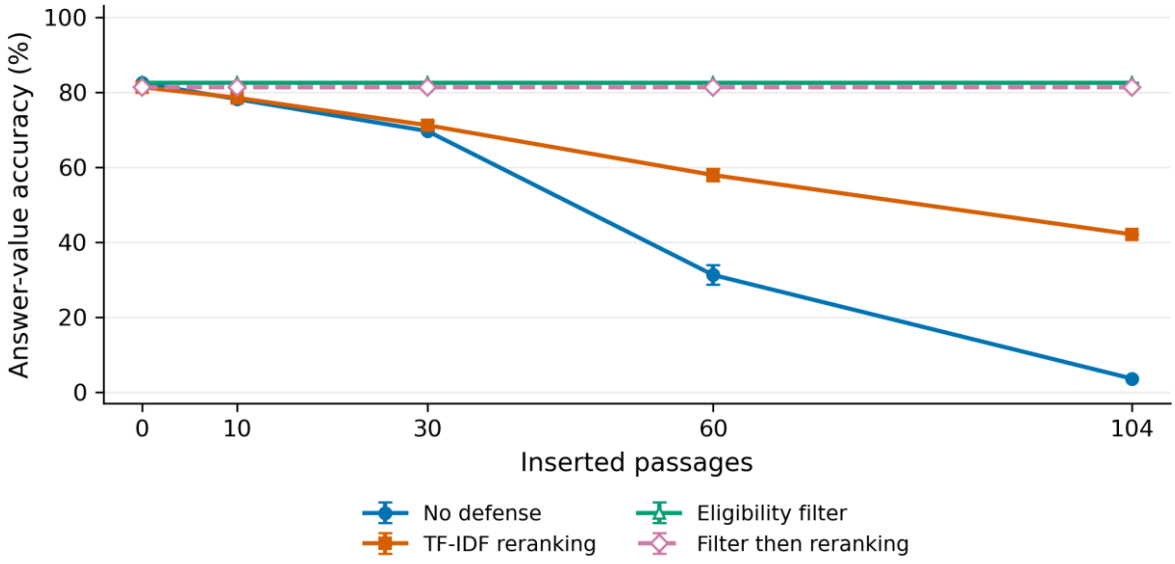


Figure 2. Answer-value accuracy across all 364 questions. Points are means over seeds 17, 42, and 91; error bars show sample standard deviations, not confidence intervals. Eligibility filtering retains its clean baseline; filtering then reranking retains the reranked baseline. Values describe deterministic extractive responses under lexical scoring.

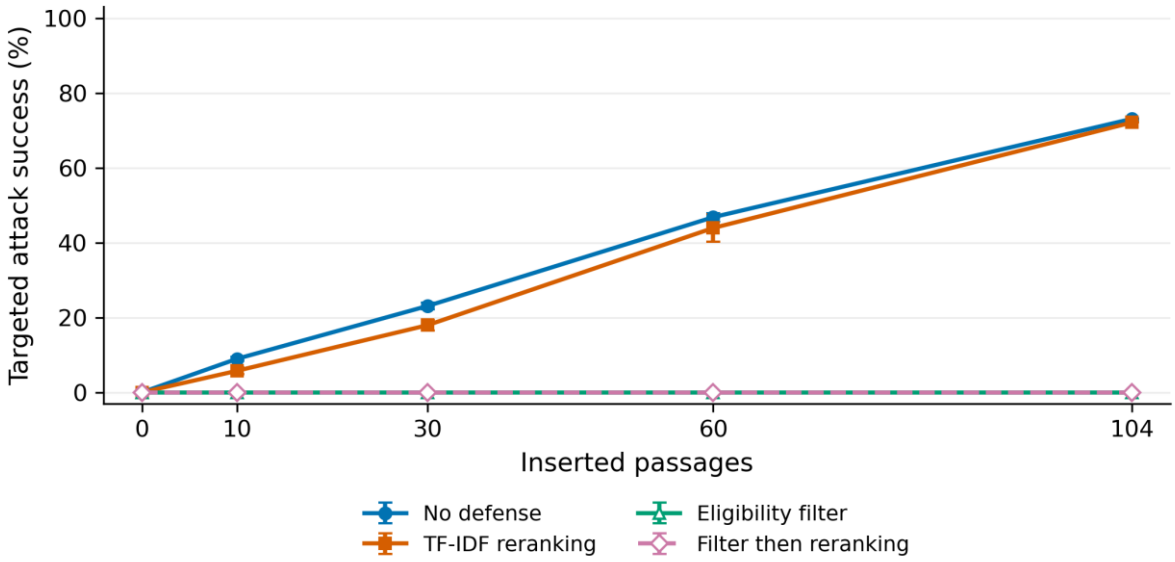


Figure 3. Target-specific, exposure-confirmed attack success across 104 designated targets. Points average the same three seeds; error bars show sample standard deviations. The two filtered curves coincide at zero under rejected attack labels. Increasing insertion count also increases the number of attacked targets; these percentages are not complements of Figure 2.

The accuracy curves measure overall utility, whereas the success curves measure intended target manipulation. At full budget, reranking raises answer-value accuracy from 3.6% to 42.0% but leaves ASR at 72.1%, compared with 73.1% without defense. Its clean accuracy is 81.3%. A preserved filtered baseline does not imply correction of clean errors, and neither figure estimates generative LLM response probabilities.

5.4. Clean-performance cost

On the clean corpus, reranking raises gold Recall@5 from 340/364 to 357/364 while answer-value accuracy decreases from 300/364 to 296/364. The net clean cost is 1.1 percentage points. This

observed decrease is consistent with H5 within the pilot; no significance or population-level effect is claimed.

Filtering preserves clean answer-value accuracy at 82.4%; filtering then reranking preserves the reranked baseline at 81.3%. The filter removes two low-trust clean chunks out of 225, but this benchmark shows no net clean answer-accuracy loss attributable to filtering. Zero observed loss is therefore a property of the supplied questions and labels, not proof that legitimate information is never rejected or that source review is cost-free. False rejection, source-review effort, latency, and memory costs require a separately designed operational evaluation.

5.5. Repetition and forged-label controls

In the separate target-specific repetition experiment, undefended successes are 90/104 with one malicious copy and 91/104 with two or more. All 104 answer strings remain unchanged between five, ten, twenty, and fifty copies. The plateau is consistent with saturation of a five-slot context; it does not imply that nominal insertion counts determine continuously worsening answers.

Reranked successes increase from 37/104 with one copy to 61/104 with two and 92/104 with five or more. Exact duplicate collapse retains 90/104 successes at every positive nominal dose: it reduces identical multiplicity to one effective passage without making that passage reliable. The intervention concerns exact repeated text, not diverse or semantically similar poisons.

At 50 identical copies per target, filtering yields 0/104 scored successes under rejected metadata; changing only those entries to high trust and verified yields 91/104 successes (87.5%) in the matched target-specific control. Each target has its own KB, rather than 5,200 copies sharing one collection. This result identifies dependence on metadata authority under changed attacker capability; it is not a direct comparison with the main shared-KB 76/104 result or an implemented authentication test. Only saved-response rescoring, not a new replay of this supplementary experiment, was performed. Table 6 keeps the protocols separate.

Table 6. Selected supplementary outcomes. Each condition evaluates 104 targets. Identical-copy conditions are target-specific and are not directly comparable to the shared-corpus budget in Table 4.

Condition	Copies per target	Successful attacks
No defense	1	90/104
No defense	2	91/104
No defense	5–50	91/104 at every measured dose
Reranking	1	37/104
Reranking	2	61/104
Reranking	5–50	92/104 at every measured dose
Exact duplicate collapse	1–50	90/104 at every positive measured dose
Eligibility filter; rejected attack labels	50	0/104
Eligibility filter; forged accepted labels	50	91/104

6. Discussion and institutional implications

6.1. Recovering evidence is an intermediate security outcome

The principal finding is a separation between evidence recovery and answer integrity. Reranking can recover the annotated supporting passage while keeping the malicious passage in context. Its large Recall@5 improvement would therefore overstate protection if considered without targeted outputs. Overall answer improvement also masks persistent vulnerability among the attacked targets. Conversely, zero scored attacks after filtering does not imply perfect correctness, because clean answer errors remain.

These distinctions support reporting a sequence of outcomes: supporting-evidence availability, malicious exposure, intended false output, and correct-answer restoration. No single measure captures the sequence. This reporting approach is especially useful when evaluating an assistant whose sources may contain competing assertions rather than explicit malicious instructions.

6.2. Trust labels represent governance assumptions

The eligibility policy is effective because its accepted/rejected labels separate malicious insertions from useful clean material. Its security boundary is therefore the authority that assigns and protects those labels. A trusted source may still be outdated or factually wrong, and compromised accepted metadata changes the threat model. Source authentication and claim validation address related but distinct problems.

The forged-label control motivates careful accounting of where trust enters a pipeline. A provenance gate should be described by its authority, verification process, update rules, and attacker access, rather than by a label name alone. The present experiment measures neither authentication implementation nor the real-world cost of source review.

6.3. Implications for institutional knowledge management

For prospective institutional use, the results support evaluating source governance and answer integrity alongside retrieval quality. Table 7 presents proposed actions tied to the observed mechanisms. These recommendations do not constitute ministry approval, evidence of deployment, or measured operational effectiveness. An institutional pilot would need accountable ownership of ingestion decisions, verifiable changes to source and trust metadata, and an answer-review process suitable for the actual user tasks.

Table 7. Proposed institutional actions and the limits of their supporting evidence.

Function	Proposed action	Research basis and boundary
Ingestion	Assign accountable source ownership and review corpus additions	Base filtering works under authoritative labels; review effectiveness is unmeasured
Provenance	Protect label changes and retain source identity, dates, and versions	Forged accepted labels remove the observed protection
Quality assurance	Evaluate answers as well as retrieval; check conflicting claims	High gold recall coexists with targeted false answers
Monitoring	Retain query, context, ranking, output, and decision records	Paired outcomes require response-level evidence

Function	Proposed action	Research basis and boundary
Incident response	Quarantine suspect entries and identify affected questions	Shared-corpus contamination reaches beyond selected targets
Recovery	Restore a clean snapshot and reassess both clean and attacked questions	Preventing an attack does not necessarily restore correctness
Deployment assessment	Verify actual model backends and measure retained utility	Offline extractive findings cannot quantify deployed LLM behavior

The proposed actions apply to corpus administration and evaluation design. The research does not justify autonomous high-stakes decisions or quantify effects on a particular public service. An institutional deployment assessment would require its actual documents, source governance, user tasks, and generative configuration.

7. Validity limitations and future research

7.1. Scope of the empirical evidence

The principal factorial study contains no generative LLM inference. Hashing and deterministic extraction do not capture neural retrieval, parametric knowledge, instruction following, stochastic generation or semantic handling of conflicting evidence. The historical answerer ignores the configured system prompt. Its numerical findings characterize that pipeline. Subsequent neural and generative development checks in Appendix D have distinct protocols and no independent human adjudication; they do not establish the original ASR for an LLM.

The small corpus is dominated by templated statistical observations. Correlated years and question formulations limit independent-sample interpretation, and inherited references require further checking. A single annotated gold chunk can omit alternative support. All existing splits were inspected, preventing claims of held-out generalization. The severe full-pool contamination rate and the absence of adaptive attack optimization also limit external validity.

7.2. Measurement and comparison limitations

Lexical scoring may misclassify negation, quotation, temporal mismatches, and mixed explanations. The success definition confirms a false-only lexical outcome with target-specific exposure; it is not a complete semantic safety measure. Strict exact match is poorly aligned with the answer format. Future evaluation should distinguish correct answers, intended false answers, other errors, conflicting claims, abstentions, and execution failures.

The dose curve combines target coverage with competition. Repeated-copy and shared-corpus experiments differ in their surrounding attack sets and must not be pooled. The present study does not independently identify effects of corpus size, context length, attack diversity, or correct-evidence density. Across-seed SD measures selection sensitivity; zero endpoint variation is not population certainty. No untouched test set, external replication, or permanent artifact identifier is claimed.

7.3. Research continuation

The final generative evaluation must freeze retriever and generator revisions, prompts, generation parameters, defense order, attack protocol and scoring rules before test outcomes are inspected. The neural components and generator have passed separate development integration checks (Appendix

D), but fresh data and independent semantic assessment remain outstanding. Failures must stop execution or create separately labelled conditions, without silent model substitution. Partition fresh questions by underlying fact and linked sources; keep related formulations together.

Matched unrelated poison and benign insertions are needed to test H1 and separate targeted effects from generic index competition. A fixed-target design should report insertion coverage, exposure among attacked targets, and success conditional on exposure. The pre-retrieval gate should be compared with a separately specified rerank-then-filter condition, including context backfilling and false rejection of legitimate documents.

Blinded human adjudication should check intended false claims and factual correctness, with independent second review and documented disagreement resolution. Numeric answers require entity, relation, year, units, and tolerance. Uncertainty should account for shared facts and source families rather than treating deterministic repetitions as independent samples. An additional QA dataset and additional generative models would strengthen external validity, but are extensions beyond the completed pilot.

8. Conclusion

This research analyzes targeted knowledge-base poisoning through a controlled offline study supported by prior literature and verified local evidence. Retrieval recovery, attack prevention, and correctness restoration are distinct outcomes: reranking recovers much of the annotated evidence while leaving most full-budget target manipulations successful. An authoritative metadata gate prevents rejected-label attacks, but accepted-label forgery removes that protection in a separate control. Repeated-copy experiments show that nominal insertion volume can exceed the effective five-passage context without changing the answer.

The study offers a bounded empirical basis for institutional evaluation of knowledge integrity. Its principal response-level results were recomputed and locally reproduced with checksum-matched historical code. That verification strengthens reproducibility within the pilot; it does not establish semantic correctness, independent-source validation of inherited facts, or general robustness of generative RAG. Further scientific claims should follow fresh evaluation data, blinded adjudication, and explicitly versioned neural and generative experiments.

Data availability and reproducibility

The local research artifact contains the archived principal run 20260921T061151Z_d1966106 (21,840 evaluations), the separately archived repetition and forged-label control (3,016 evaluations), and the revised-backend diagnostic (1,092 evaluations). All three sets were inspected at response level and their reported counts recomputed. Only the principal grid was fully re-executed in the present verification. Datasets, manifests, exact historical source, independent audit scripts, recomputed tables, and two independent plotting outputs are retained with the revised manuscript. Counts refer to evaluations, not independent samples, and protocols are not pooled. Subsequent development archives retain 1,120 neural retrieval records, 24 neural/generative responses and 1,344 fixed-target retrieval-control records, with separate manifests and analysis boundaries (Appendix D). These counts are not added to the principal sample size.

No public repository, permanent artifact identifier, or external replication is claimed. The main archived results.csv has SHA-256 349c3bcfd7847843c44bd745111e2872d23cd28e26de4ff2361f6543bad485cf. Reproduction requires the frozen data and source hashes, original backend settings and budgets, dependency versions, attack selections, scoring implementation, and response records. The current application's defaults and later

response-selection code are not a substitute for the archived experiment. A public release remains subject to provenance and licensing review.

Responsible research

Experiments concern a local research corpus. Artificial false passages must remain identified as attack material and separated from factual source records. Any artifact release requires provenance and licensing review and removal of private credentials. The research does not require modifying public sources or attacking third-party systems.

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Appendix A. Relationship to the original research plan

The original proposal document was not supplied for independent inspection. The following comparison therefore concerns the research plan as described in earlier manuscripts, rather than confirmed compliance with a separately reviewed proposal. That description anticipated approximately 100-150 documents, 500-1,500 chunks, 200-300 questions, and 50 targets. The executed corpus instead contains 224 documents, 225 chunks, 364 questions, and 104 targets. Many records are short statistical observations, so document count does not measure linguistic diversity.

The described plan involved ChromaDB, neural embeddings, a fixed generative LLM, and filtering after reranking. The verified principal run instead uses exact cosine search, token hashing, deterministic extraction, and eligibility before retrieval. Its budgets are 0, 10, 30, 60, and 104, compared with described planned budgets of 0, 5, 10, and 20. Current application defaults use 0, 10, 20, 50, and 104 and must not replace the principal run's configuration. These are disclosed implementation and protocol deviations; the planned neural and generative components were not executed in the principal study. Subsequent development checks are reported separately in Appendix D.

Appendix B. Later diagnostic and version sensitivity

The diagnostic of 30 September 2026 uses hashing-lexical-v2:1024d retrieval with sentence-coverage heuristics and an extractive-evidence-v2 answerer. All 364 benchmark questions receive synthetic QA assertions, including questions untargeted in the principal 104-target pool. Each question is evaluated in a separate KB with either zero or ten identical copies of its own assertion. Complete response records confirm 355/364 clean answer matches, 337/364 with ten copies and no defense, and 355/364 with filtering. There are 27 scored target successes without defense and zero with filtering. The diagnostic contains 1,092 evaluations.

This diagnostic changes retrieval, sentence selection, attack wording, and the target population. Its 27/364 success count (7.4%) consequently has a different denominator and protocol from principal

ASR. It includes successes on some questions already wrong at clean baseline and cannot be equated with the net loss of 18 answer matches. The filtered answers match their corresponding clean answers exactly. Results are retained as a separate implementation diagnostic; they are not a replication of the principal grid, a real-LLM experiment, or an estimate of a single isolated software change.

Appendix C Evidence verification and reproducibility

Local verification separates original measurements, recomputation of saved responses, and historical pipeline re-execution. Evidence copies and checksum-matched modules are retained separately from the current application. A later evaluator export line was removed only after the resulting complete file matched the main manifest's original SHA-256.

Table 8. Verification scope and remaining boundaries.

Evidence	Completed verification	Boundary
Principal archive	5 CSV hashes matched; 21,840 unique records in 60 groups	Manifest consistency; no external certification
Reference data	4 input hashes; 168 WDI observations and 336 reference values matched	Inherited documents not independently fact-checked
Scoring and retrieval	Every main lexical verdict and saved context membership checked	Value containment remains a heuristic
Historical re-execution	32 checksum-matched modules; 21,840 outcomes replayed	Local execution; latency excluded
Repetition and label forgery	3,016 records rescored; result and protocol hashes matched	Separate protocol; no new replay
Later diagnostic	1,092 records; ten own-assertion copies per attacked question confirmed	Different backend, attacks, and target set

The replay used Python 3.13.9, NumPy 2.3.5, pandas 2.3.3, and scikit-learn 1.7.2. Answer strings, retrieved IDs and rounded scores, exposure and scoring flags, and token F1 matched all 21,840 archived records. No model download or API generation was used. Timing and raw output-byte equivalence are excluded.

Verification scripts and derived tables are retained in research/audit, including audit_evidence.py, reproduce_historical.py, and replication/replication_report.json. Archived evidence and historical code are in research/evidence. The independent plots are Figure_2_Answer_Accuracy.png and Figure_3_Attack_Success.png. These are local artifact paths; no public archive or institutional endorsement is claimed.

Appendix D Subsequent development evidence

This appendix records development work completed on 6 October 2026. None of these checks is an untouched final test, an adaptive attack search, independent human adjudication or an external replication. The historical numerical results remain attributed to their original backend.

D1 Pinned neural retrieval and generator integration

Run 20261006T064427Z_4cf05372 completed 1,120 retrieval-only records on 140 previously seen development questions with 48 targets and four defenses. Local all-MiniLM-L6-v2 embeddings use revision 1110a243fdf4706b3f48f1d95db1a4f5529b4d41; the MS MARCO MiniLM cross-encoder

uses revision 233902d25c440f23af6f7d6e94d2946bac0bee0a. Weight hashes and runtime versions are retained. CPU exact cosine retrieval selects 30 candidates and five final passages; eligibility filtering precedes retrieval. No answer generation occurred in that run.

Run 20261006T070136Z_e3ed4898 used frozen neural contexts with gpt-4.1-mini-2025-04-14. It completed 24 responses: three development agricultural-land indicator-year facts (2010, 2012, 2014), clean versus three total malicious insertions, and four defenses. Requests used temperature 0, at most 300 output tokens and no fallback. All returned model snapshots, payload/output hashes and usage totals passed audit. Temperature 0 is not a determinism guarantee. Usage was 10,119 input and 927 output tokens; the estimate at recorded standard uncached rates is USD 0.0055308, not a verified invoice.

All 24 final contexts contained the reference chunk. The target poison also appeared in the six attacked unfiltered contexts, including the three reranked contexts. One answer mentions the correct 58.13 but gives priority to 48.25 as a revised value. This illustrative response motivates endorsement-aware scoring: mentioning a correct number does not establish correctness. No semantic accuracy or ASR is estimated from these three correlated facts. Two independent review templates exist, but zero independent human judgments are completed.

D2 Fixed-target retrieval controls

Run 20261006T071953Z_ee1839e0 completed 1,344 retrieval-only records for a fixed set of 48 previously seen development targets. The seven corpus conditions are clean and one, two or five exact copies per target, under rejected metadata or accepted-metadata compromise. All positive doses cover every target. Corresponding rejected/accepted conditions change only trust and verification labels, keeping assertion text and insertion count identical. Four defense conditions retain 30 candidates and five final passages. Copy doses produce 48, 96 or 240 insertions; the resulting contamination shares are 17.6%, 29.9% and 51.6%. These remain stress conditions rather than evidence about low operational contamination.

At five copies per target with source filtering, own-poison exposure is 0/48 under rejected labels and 43/48 under accepted-label compromise. These are final-context inclusion counts, not successful answers. Saved insertion files, fixed coverage, paired metadata, output hashes and every retrieval indicator were checked. No generator was called and no ASR, semantic correctness or H1 comparison is established. Exact repetition does not implement the proposed diverse-poison dose experiment.

D3 Final evaluation work remaining

Two source-traceable drafts contain 82 candidate facts and 164 question formulations. Fourteen heritage candidates remain development-only because their family was seen; 68 others await source, date, unit, question-quality and overlap review. None is frozen or test-eligible. Additional source-family diversity, matched benign and unrelated controls, independently checked references, a frozen final generative protocol and independent semantic adjudication remain necessary. Additional models and adaptive optimization are separately specified extensions; claims must follow the capabilities and conditions actually tested.

Development evidence and audit reports are retained in research/continuation: NEURAL_DEVELOPMENT_STATUS.json, NEURAL_GENERATIVE_PILOT_STATUS.json, FIXED_TARGET_CONTROL_STATUS.json and DATASET_EXPANSION_STATUS.json. A source/configuration/input snapshot accompanies the generative pilot. Review templates have not been substituted for labels. No public repository, formal institutional approval or journal acceptance is claimed.